

Analytical Gradient Approximation of Affirmation Loss for Efficient Token Attribution*

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1 Method Overview and Affirmation Loss

The Token Highlighter ([see original paper](#)) inspects and mitigates adversarial jailbreak threats in user queries by computing per-token influence scores using the ℓ_2 -norm of the input gradients, $\left\| \frac{\partial \mathcal{L}}{\partial \mathbf{x}_i} \right\|_2$. The **Affirmation Loss**, $\mathcal{L}$, evaluates the negative log-likelihood of a predefined target phrase $\mathbf{y}$ conditioned on a user query with token embeddings $\mathbf{x}_i$. The method applies soft removal by shrinking the embeddings of ‘driver tokens’ encouraging model compliance.

In production serving systems such as vLLM, executing a standard full autograd backward pass during inference introduces prohibitive latency and memory overhead, as intermediate activations are discarded to optimize paged memory layouts. Our analytical formulation avoids this bottleneck. It requires only a single forward pass through the prefill stage to extract attention activations and weight matrices via hooks, mapping token attributions back along the sequence axis without the need for a global backpropagation graph.

Index the concatenated prompt and response tokens as the set $R = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n, \mathbf{j}_0, \mathbf{j}_1, \dots, \mathbf{j}_m\}$, where prompt tokens are 1-indexed and generated target tokens are 0-indexed. Define $t = n + j$ as the absolute sequence position of the j -th generated token, where the target generation window spans $t \in T = \{n, n + 1, \dots, n + m - 1\}$. Let $\mathbf{z}_t = W_U \mathbf{h}_t^{(L)}$ be the logit vector at step t , where $W_U \in \mathbb{R}^{V \times d_{\text{model}}}$ is the unembedding matrix. The affirmation loss is the teacher-forced negative log-likelihood over the target affirmative sequence $\mathbf{y}$:

$$\mathcal{L}(\mathbf{x}_{1:n}) = - \sum_{t \in T} \log P_{\theta}(y_{t-n} \mid \mathbf{x}_{1:n}, y_{<t-n}) = - \sum_{t \in T} \log \sigma(\mathbf{z}_t)_{y_{t-n}}, \quad (1)$$

where $\sigma(\cdot)$ is the softmax operator and $\sigma(\mathbf{z}_t)_{y_{t-n}}$ is the scalar probability assigned to the ground-truth affirmative target token at step t .

2 Residual Stream Approximation and Layer Topologies

In an L -layer decoder-only transformer, the input embedding $\mathbf{x}_i$ initializes the residual stream at sequence position i . For the final transformer block, the hidden state vector accumulates the

*This work is part of an ongoing collaboration with Pin-Yu Chen and Irene Ko at IBM Research.

attention and feed-forward network (FFN) contributions sequentially. Let $\mathbf{h}_t^{(L-1)}$ be the hidden state entering the final block. The forward progression is:

$$\mathbf{h}_{mid,t} = \mathbf{h}_t^{(L-1)} + \text{Attn}(\text{Norm}_1(\mathbf{h}_t^{(L-1)})), \quad (2)$$

$$\mathbf{h}_t^{(L)} = \mathbf{h}_{mid,t} + \text{FFN}(\text{Norm}_2(\mathbf{h}_{mid,t})). \quad (3)$$

Following the final residual addition, the stream passes through a terminal bottleneck normalization layer before projecting to the vocabulary logits via the Language Modeling (LM) head:

$$\hat{\mathbf{h}}_t^{(L)} = \text{Norm}_f(\mathbf{h}_t^{(L)}), \quad (4)$$

$$\mathbf{z}_t = W_U \hat{\mathbf{h}}_t^{(L)}, \quad (5)$$

where $\hat{\mathbf{h}}_t^{(L)}$ is the normalized final hidden state and $W_U \in \mathbb{R}^{V \times d_{\text{model}}}$ is the unembedding weight matrix.

During backpropagation, the boundary error signal $\mathbf{g}_{out}$ evaluated at input to the must be projected back through this terminal normalization layer. Expanding via the chain rule incorporates the Jacobian of the final norm, $J_{\text{Norm}_f}^\top$, mapping the downstream logit gradient to the block output:

$$\mathbf{g}_{out} = \frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}} = J_{\text{Norm}_f}^\top \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{h}}_t^{(L)}} = J_{\text{Norm}_f}^\top W_U^\top \frac{\partial \mathcal{L}}{\partial \mathbf{z}_t}. \quad (6)$$

Once $\mathbf{g}_{out}$ is isolated, it splits at each preceding addition node in the backward pass. By the multivariate Chain Rule, the exact total derivative through this final block expands into the following product of Jacobians:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{h}_i^{(L-1)}} = \left(\frac{\partial \mathbf{h}_{mid,t}}{\partial \mathbf{h}_i^{(L-1)}} \right)^\top \left(\frac{\partial \mathbf{h}_t^{(L)}}{\partial \mathbf{h}_{mid,t}} \right)^\top \frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}} = \left(I + J_{\text{Norm}_1}^\top J_{\text{Attn}}^\top \right) \left(I + J_{\text{Norm}_2}^\top J_{\text{FFN}}^\top \right) \mathbf{g}_{out}. \quad (7)$$

Because the token vector $\mathbf{x}_i$ propagates through prior successive skip-connections, the analytical Jacobian $\frac{\partial \mathbf{h}_i^{(L-1)}}{\partial \mathbf{x}_i}$ expands to the identity matrix I plus cross-token feedback. **Assuming that direct identity paths dominate local sensitivity in prior layers, we omit early cross-token feedback components** and approximate the transformation boundary as:

$$\frac{\partial \mathbf{h}_i^{(L-1)}}{\partial \mathbf{x}_i} \approx I \implies \frac{\partial \mathcal{L}}{\partial \mathbf{x}_i} = \frac{\partial \mathcal{L}}{\partial \mathbf{h}_i^{(L-1)}} \cdot \frac{\partial \mathbf{h}_i^{(L-1)}}{\partial \mathbf{x}_i} \approx \frac{\partial \mathcal{L}}{\partial \mathbf{h}_i^{(L-1)}}.$$

We visualize the computational graph as follows:

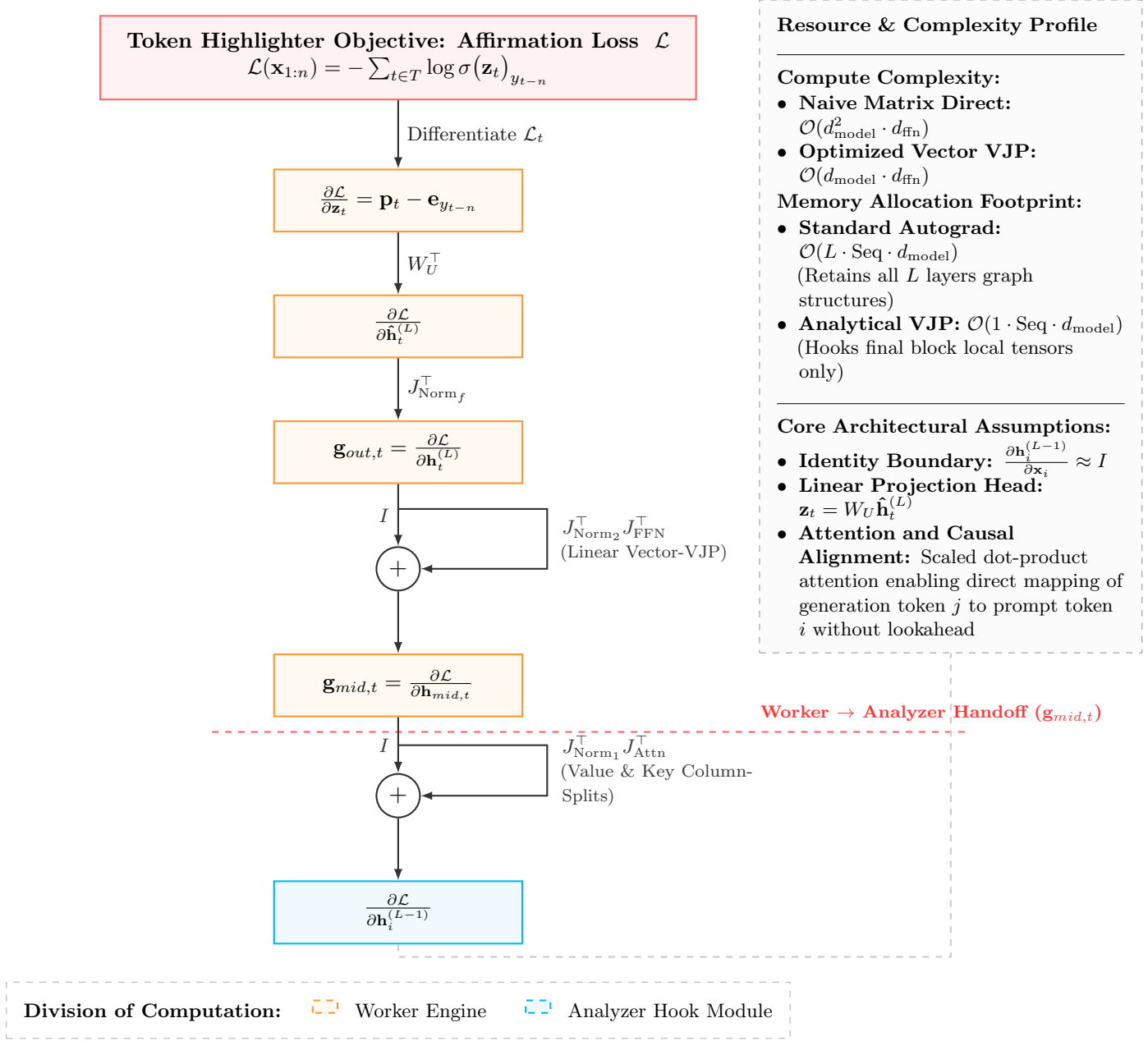


Figure 1: Execution topology showing system module division between the host vLLM Worker processing engine and the Token Highlighter Analyzer pipeline. Memory-intensive computations are handled by the worker while large tensors are actively present, and smaller vector results are passed to the analyzer. Analytical optimizations compress backward operations down to simple vector operations, avoiding expensive graph allocation. The affirmation gradient signal is computed with respect to the normalized final-layer hidden states and backprojected through the final normalization layer ($\mathbf{g}_{\text{out},t}$) before moving through the FFN module ($\mathbf{g}_{\text{mid},t}$) and attention module to end at the beginning of the last decoder block.

To compute Equation (7) exactly for the final layer, we derive the boundary gradient $\mathbf{g}_{out}$, followed by the independent analytical VJPs for the Attention and FFN sub-layers.

Remark 1 (Architectural Scope and Approximations). *This derivation explicitly assumes a standard decoder-only autoregressive transformer layout. This assumption provides three operational properties:*

- (i) *Prompt tokens and generated text share a unified sequence axis, allowing cross-token dependencies to be mapped via a single causal self-attention matrix.*
- (ii) *The language modeling head is a linear projection W_U , enabling exact analytical isolation of the boundary gradient.*
- (iii) *The underlying software runtime supports the extraction of attention activations $(\alpha, \mathbf{q}, \mathbf{k}, \mathbf{v})$ and block inputs via forward-pass hooks.*

3 Gradient of the Language Modeling (LM) Head

We isolate the loss gradient with respect to the final hidden state at step t , denoted $\mathbf{g}_{out} = \frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}}$. Expanding via the multivariate Chain Rule across the concatenated sequence yields:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}} = J_{\text{Norm}_f}^\top \left(\underbrace{\sum_{s=0}^{n-1} \frac{\partial \mathcal{L}}{\partial \mathbf{z}_s} \frac{\partial \mathbf{z}_s}{\partial \mathbf{h}_t^{(L)}}}_{\text{Prompt Tokens (0)}} + \underbrace{\sum_{s=n}^{n+m-1} \frac{\partial \mathcal{L}}{\partial \mathbf{z}_s} \frac{\partial \mathbf{z}_s}{\partial \mathbf{h}_t^{(L)}}}_{\text{Generated Tokens}} \right) = J_{\text{Norm}_f}^\top \frac{\partial \mathcal{L}}{\partial \mathbf{z}_t} \frac{\partial \mathbf{z}_t}{\partial \mathbf{h}_t^{(L)}}. \quad (8)$$

The prompt-token summation vanishes because the affirmation loss does not consider logits evaluated during the prefill phase ($\frac{\partial \mathcal{L}}{\partial \mathbf{z}_s} = \mathbf{0}$ for $s < n$). Since the logit vector $\mathbf{z}_s$ depends strictly on the corresponding hidden state ($\mathbf{z}_t = W_U \hat{\mathbf{h}}_t^{(L)}$), the cross-position derivative is zero for all $s \neq t$.

Differentiating the scalar loss contribution $\mathcal{L}_t = -\log \sigma(\mathbf{z}_t)_{y_{t-n}}$ with respect to an arbitrary logit element $z_{t,k}$ and applying the softmax Jacobian identity $\frac{\partial \sigma(\mathbf{z})_k}{\partial z_r} = \sigma(z_k)(\delta_{kr} - \sigma(z_r))$ yields the coordinate derivative:

$$\frac{\partial \mathcal{L}_t}{\partial z_{t,k}} = -\frac{1}{\sigma(\mathbf{z}_t)_{y_{t-n}}} \cdot [\sigma(\mathbf{z}_t)_{y_{t-n}} (\delta_{y_{t-n},k} - \sigma(\mathbf{z}_t)_k)] = p_{t,k} - \delta_{y_{t-n},k}.$$

Letting $\mathbf{p}_t = \sigma(\mathbf{z}_t) \in \mathbb{R}^V$ denote the dense vocabulary probability vector, we collect these derivatives:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{z}_t} = \mathbf{p}_t - \mathbf{e}_{y_{t-n}}, \quad (9)$$

where $\mathbf{e}_{y_{t-n}} \in \mathbb{R}^V$ is the standard one-hot basis vector. Given $\mathbf{z}_t = W_U \mathbf{h}_t^{(L)}$, the sensitivity matrix is W_U . Substituting these terms determines the exact boundary condition:

$$\mathbf{g}_{out,t} = \frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}} = J_{\text{Norm}_f}^\top W_U^\top (\mathbf{p}_t - \mathbf{e}_{y_{t-n}}). \quad (10)$$

4 Attention Mechanism: Key, Value and Query Paths

From Equation (7), let $\mathbf{g}_{mid,t} = (I + J_{\text{Norm}_2}^\top J_{\text{FFN}}^\top) \mathbf{g}_{out,t}$ be the intermediate gradient vector entering the attention addition node. We back-project $\mathbf{g}_{mid}$ through the final causal attention block to uncover direct and indirect effects of prompt token $\mathbf{x}_i$:

1. **Value Path:** Tracks semantic magnitude, mapping how updates in the representation of prompt token $\mathbf{x}_i$ propagate features directly into the residual stream via the value vector.
2. **Key Path:** Tracks routing variations, measuring how changes in the key vector of prompt token $\mathbf{x}_i$ modulate the causal attention weights.
3. **Query Path:** Tracks how modification to the query vector of prompt token $\mathbf{x}_i$ influences the affirmation loss via the first generation token (only non-zero for boundary token $\mathbf{x}_n$).

Within the attention block, the update at position t expands as:

$$\mathbf{h}_{mid,t} = \underbrace{\mathbf{h}_t^{(L-1)}}_{\text{Skip Connection}} + \underbrace{W_O \sum_{k=0}^t \alpha_{tk} \mathbf{v}_k}_{\text{Attention Output}}.$$

Applying the product rule to the prompt index column $k = i$ isolates the total Jacobian of the raw attention context vector $\mathbf{o}_t$:

$$\frac{\partial \mathbf{o}_t}{\partial \mathbf{h}_i^{(L-1)}} = \underbrace{\alpha_{ti} W_V}_{\text{Value Path}} + \underbrace{\sum_{k=0}^t \mathbf{v}_k \left(\frac{\partial \alpha_{tk}}{\partial \mathbf{h}_i^{(L-1)}} \right)^\top}_{\text{Key Path}}.$$

4.1 Value Path Contribution

Holding α_{ti} fixed, the structural mapping reduces to the linear operator $W_O \alpha_{ti} W_V$. Projecting the intermediate gradient $\mathbf{g}_{mid,t}$ backward across the generation window yields:

$$\left[J_{\text{Attn}}^\top \mathbf{g}_{mid} \right]_{i,\text{value}} = \sum_{t \in T} (W_O \alpha_{ti} W_V)^\top \mathbf{g}_{mid,t} = W_V^\top W_O^\top \sum_{t \in T} \alpha_{ti} \mathbf{g}_{mid,t}. \quad (11)$$

4.2 Key Path Contribution

Let $\mathbf{q}_t$, $\mathbf{k}_i$, and $\mathbf{v}_i$ denote the query, key, and value vectors, where the attention score is $s_{ti} = \frac{\mathbf{q}_t^\top \mathbf{k}_i}{\sqrt{d}}$ and $\alpha_{ti} = \sigma(s_{ti})$. Differentiating the context vector $\mathbf{o}_t$ relative to s_{ti} yields:

$$\frac{\partial \mathbf{o}_t}{\partial s_{ti}} = \sum_{j \leq t} \mathbf{v}_j \frac{\partial \alpha_{tj}}{\partial s_{ti}} = \sum_{j \leq t} \mathbf{v}_j \alpha_{tj} (\delta_{ji} - \alpha_{ti}) = \alpha_{ti} (\mathbf{v}_i - \bar{\mathbf{o}}_t),$$

where $\bar{\mathbf{o}}_t = \sum_{j \leq t} \alpha_{tj} \mathbf{v}_j$ is the attention-weighted mean value vector at step t .

Mapping the sensitivity of the loss function to s_{ti} :

$$\frac{\partial \mathcal{L}}{\partial s_{ti}} = \mathbf{g}_{mid,t}^\top W_O \frac{\partial \mathbf{o}_t}{\partial s_{ti}} = \mathbf{g}_{mid,t}^\top W_O \alpha_{ti} (\mathbf{v}_i - \bar{\mathbf{o}}_t).$$

Define the scalar alignment score $g_{ti} := \mathbf{g}_{mid,t}^\top W_O \mathbf{v}_i$ and baseline $\bar{g}_t := \sum_{j \leq t} \alpha_{tj} g_{tj}$. The sensitivity reduces to the centered excess alignment:

$$\frac{\partial \mathcal{L}}{\partial s_{ti}} = \alpha_{ti} (g_{ti} - \bar{g}_t).$$

Given $\frac{\partial s_{ti}}{\partial \mathbf{h}_i^{(L-1)}} = \frac{W_K^\top \mathbf{q}_t}{\sqrt{d}}$, the key path metric evaluates to:

$$\left[J_{\text{Attn}}^\top \mathbf{g}_{mid} \right]_{i,\text{key}} = \frac{W_K^\top}{\sqrt{d}} \sum_{t \in T} \alpha_{ti} (g_{ti} - \bar{g}_t) \mathbf{q}_t. \quad (12)$$

4.3 Boundary Constraints and the Query Path

Reconciling this analytical framework with production inference environments (such as vLLM) that employ KV-caching requires two structural boundary conditions. These arise because the loss evaluation window sits strictly within the generated response ($t \in T \implies t \geq n+1$), where n denotes the prompt length.

1. **Query Path Discontinuity:** Evaluating the full attention score derivative requires tracking how a perturbation to input token $\mathbf{h}_i^{(L-1)}$ affects the score at an arbitrary generation step t :

$$\frac{\partial s_{ti}}{\partial \mathbf{h}_i^{(L-1)}} = \delta_{ti} \frac{W_Q^\top \mathbf{k}}{\sqrt{d}} + \frac{W_K^\top \mathbf{q}_t}{\sqrt{d}}.$$

Because $t \geq n+1$, the query-matching condition $t = i$ is structurally impossible for any internal prompt token ($i < n$). Their query representations are never accessed during decoding, yielding $[J_{\text{Attn}}^\top \mathbf{g}_{mid}]_{i,\text{query}} = \mathbf{0}$. The query path activates exclusively at the boundary token ($i = n$) to drive the generation of the first response token ($t = n+1$). Setting $t = n+1$ and tracking its interaction with prompt keys $j \leq n+1$ isolates the boundary gradient:

$$\left[J_{\text{Attn}}^\top \mathbf{g}_{mid} \right]_{n,\text{query}} = \frac{W_Q^\top}{\sqrt{d}} \sum_{j=1}^n \alpha_{nj} (g_{nj} - \bar{g}_n) \mathbf{k}_j \quad (13)$$

2. **Skip Connection Gradient:** In global backpropagation, the total derivative of the block input receives a direct vector-Jacobian product via the residual connection, defined as $\mathbf{g}_{mid,i} = \frac{\partial \mathcal{L}}{\partial \mathbf{h}_{mid,i}}$. However, since intermediate hidden states are not retained in paged memory during KV-cached serving, the decoding graph only accesses the static K and V caches for past positions. Thus, $\mathbf{g}_{mid,i} = \mathbf{0}$ for all internal prompt tokens ($i < n$), remaining active only at the boundary ($i = n$).

Synthesizing these hardware-software co-design constraints, the total analytical gradient evaluating the prompt token representation at position i simplifies to the following conditional piecewise formulation:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{h}_i^{(L-1)}} = \begin{cases} J_{\text{Norm}_1}^\top \left([J_{\text{Attn}}^\top \mathbf{g}_{mid}]_{i,\text{value}} + [J_{\text{Attn}}^\top \mathbf{g}_{mid}]_{i,\text{key}} \right) & \text{if } i < n, \\ \mathbf{g}_{mid,n} + J_{\text{Norm}_1}^\top \sum_{p \in \{\text{value}, \text{key}, \text{query}\}} [J_{\text{Attn}}^\top \mathbf{g}_{mid}]_{n,p} & \text{if } i = n. \end{cases} \quad (14)$$

We manually add both $\mathbf{g}_{mid,n}$ and the query path term $[J_{\text{Attn}}^\top \mathbf{g}_{mid}]_{n,\text{query}}$ to the full gradient to mirror standard backpropagation behavior.

5 Feed-Forward Network (FFN) Gradients

To compute $\mathbf{g}_{mid,t}$ exactly without the memory overhead of an autograd computational graph, we evaluate the analytical Vector-Jacobian Product (VJP) of the final Feed-Forward layer, resolving $J_{\text{FFN}}^\top \mathbf{g}_{out,t}$. We generalize across the two primary architectural paradigms deployed in modern causal language models: GELU and SwiGLU (Swish activation function). Let $\mathbf{x} = \text{Norm}_2(\mathbf{h}_{mid,t})$ be the normalized residual stream entering the FFN.

5.1 Gated MLPs (SwiGLU)

Modern models (e.g., LLaMA, Qwen, Mistral) utilize a gated activation mechanism. The forward pass splits the input into parallel projections before fusing them via a Hadamard product:

$$\mathbf{g} = \mathbf{x}W_{\text{gate}}^\top, \quad \mathbf{u} = \mathbf{x}W_{\text{up}}^\top, \quad \mathbf{a} = \text{SiLU}(\mathbf{g}) \odot \mathbf{u}, \quad \mathbf{y} = \mathbf{a}W_{\text{down}}^\top.$$

The SiLU (Swish) activation is defined analytically as:

$$\text{SiLU}(\mathbf{x}) = \frac{\mathbf{x}}{2} \odot \left(\tanh\left(\frac{\mathbf{x}}{2}\right) + 1 \right). \quad (15)$$

Its exact derivative evaluates to:

$$\text{SiLU}'(\mathbf{x}) = \frac{1}{2} \left(\tanh\left(\frac{\mathbf{x}}{2}\right) + 1 \right) + \frac{\mathbf{x}}{4 \cosh^2\left(\frac{\mathbf{x}}{2}\right)} = \frac{\mathbf{x} + \sinh(\mathbf{x})}{4 \cosh^2\left(\frac{\mathbf{x}}{2}\right)} + \frac{1}{2}. \quad (16)$$

Applying the multivariate chain rule across the split branches, the exact analytical Jacobian transpose for the SwiGLU FFN resolves to:

$$J_{\text{SwiGLU}}^\top = W_{\text{up}}^\top \text{diag}(\text{SiLU}(\mathbf{g}))W_{\text{down}}^\top + W_{\text{gate}}^\top \text{diag}(\mathbf{u} \odot \text{SiLU}'(\mathbf{g}))W_{\text{down}}^\top. \quad (17)$$

However, directly instantiating $J_{\text{SwiGLU}}^\top$ creates a prohibitive memory allocation bottleneck of size $\mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$. To preserve the memory savings of the token highlighter framework, we exploit the right-associativity of the Vector-Jacobian Product. By streaming the incoming vector $\mathbf{g}_{out,t} = \frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}}$ from the right and eliminating the expensive $\text{diag}(\cdot)$ representations via element-wise multiplication ($\odot$), we rewrite the execution step to execute in linear hidden space:

$$\mathbf{d}_t = \mathbf{g}_{out,t}W_{\text{down}} \in \mathbb{R}^{1 \times d_{\text{ffn}}}, \quad (18)$$

$$\mathbf{m}_{\text{up},t} = \mathbf{d}_t \odot \text{SiLU}(\mathbf{g}_t) \in \mathbb{R}^{1 \times d_{\text{ffn}}}, \quad (19)$$

$$\mathbf{m}_{\text{gate},t} = \mathbf{d}_t \odot \mathbf{u}_t \odot \text{SiLU}'(\mathbf{g}_t) \in \mathbb{R}^{1 \times d_{\text{ffn}}}, \quad (20)$$

$$\mathbf{g}_{mid,t}^{\text{FFN}} = J_{\text{SwiGLU}}^\top \mathbf{g}_{out,t} = \mathbf{m}_{\text{up},t}W_{\text{up}} + \mathbf{m}_{\text{gate},t}W_{\text{gate}} \in \mathbb{R}^{1 \times d_{\text{model}}}. \quad (21)$$

5.2 Standard MLPs (GELU)

For traditional non-gated architectures, the forward pass simplifies to $\mathbf{g} = \mathbf{x}W_{\text{in}}^\top$, $\mathbf{a} = \Phi(\mathbf{g})$, and $\mathbf{y} = \mathbf{a}W_{\text{out}}^\top$. Using the standard tanh approximation for GELU:

$$\Phi(\mathbf{x}) = \frac{1}{2} \left(1 + \tanh \left(\sqrt{\frac{2}{\pi}} (\mathbf{x} + 0.044715\mathbf{x}^3) \right) \right). \quad (22)$$

The analytical derivative is given by:

$$\Phi'(\mathbf{x}) = 0.5(1 + \tanh(\dots)) + 0.5\mathbf{x} \odot \text{sech}^2(\dots) \sqrt{\frac{2}{\pi}}(1 + 3(0.044715)\mathbf{x}^2). \quad (23)$$

The corresponding VJP reduces to a single path:

$$J_{\text{GELU}}^\top = W_{\text{in}}^\top \text{diag}(\Phi'(\mathbf{g})) W_{\text{out}}^\top. \quad (24)$$

Mirroring the SwiGLU vectorization path, the actual computation eliminates matrix-matrix operations over the intermediate hidden layer (of dimension d_{ffn} by associating the incoming (back-projected) gradient $\mathbf{g}_{\text{out},t}$ sequentially:

$$\mathbf{d}_t = \mathbf{g}_{\text{out},t} W_{\text{out}} \in \mathbb{R}^{1 \times d_{\text{ffn}}}, \quad (25)$$

$$\mathbf{m}_{\text{in},t} = \mathbf{d}_t \odot \Phi'(\mathbf{g}_t) \in \mathbb{R}^{1 \times d_{\text{ffn}}}, \quad (26)$$

$$\mathbf{g}_{\text{mid},t}^{\text{FFN}} = J_{\text{GELU}}^\top \mathbf{g}_{\text{out},t} = \mathbf{m}_{\text{in},t} W_{\text{in}} \in \mathbb{R}^{1 \times d_{\text{model}}}. \quad (27)$$

6 Complete Analytical Gradient Formulation

By synthesizing the sequential Jacobians of the final block, the total analytical gradient evaluating the prompt token representation at position i is formalized as:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{h}_i^{(L-1)}} = \mathbf{g}_{\text{mid},i} + J_{\text{Norm}_1}^\top \left(\underbrace{W_V^\top W_O^\top \sum_{t \in T} \alpha_{ti} \mathbf{g}_{\text{mid},t}}_{\text{Value Path Magnitude}} + \underbrace{\frac{W_K^\top}{\sqrt{d}} \sum_{t \in T} \alpha_{ti} (g_{ti} - \bar{g}_t) \mathbf{q}_t}_{\text{Key Path Routing Sensitivity}} \right), \quad (28)$$

where the intermediate block vector $\mathbf{g}_{\text{mid},t} = \mathbf{g}_{\text{out},t} + J_{\text{Norm}_2}^\top J_{\text{FFN}}^\top \mathbf{g}_{\text{out},t}$ explicitly incorporates the closed-form FFN gradient ($J_{\text{SwiGLU}}^\top$ or $J_{\text{GELU}}^\top$) derived in the prior section.

The **value path** isolates direct semantic contribution, while the **key path** captures attention routing updates. The separate term $\mathbf{g}_{\text{mid},i}$ accounts for gradient signal that propagates through the FFN after bypassing the attention block. Invoking the direct residual stream identity from Section 2, the final ranking metric used to isolate and construct the critical token subset $\mathcal{Q}$ is:

$$\text{Influence}(\mathbf{x}_i) = \left\| \frac{\partial \mathcal{L}}{\partial \mathbf{h}_i^{(L-1)}} \right\|_2 \approx \left\| \frac{\partial \mathcal{L}}{\partial \mathbf{x}_i} \right\|_2.$$